

PLC-Based Automated Conveyor Control System

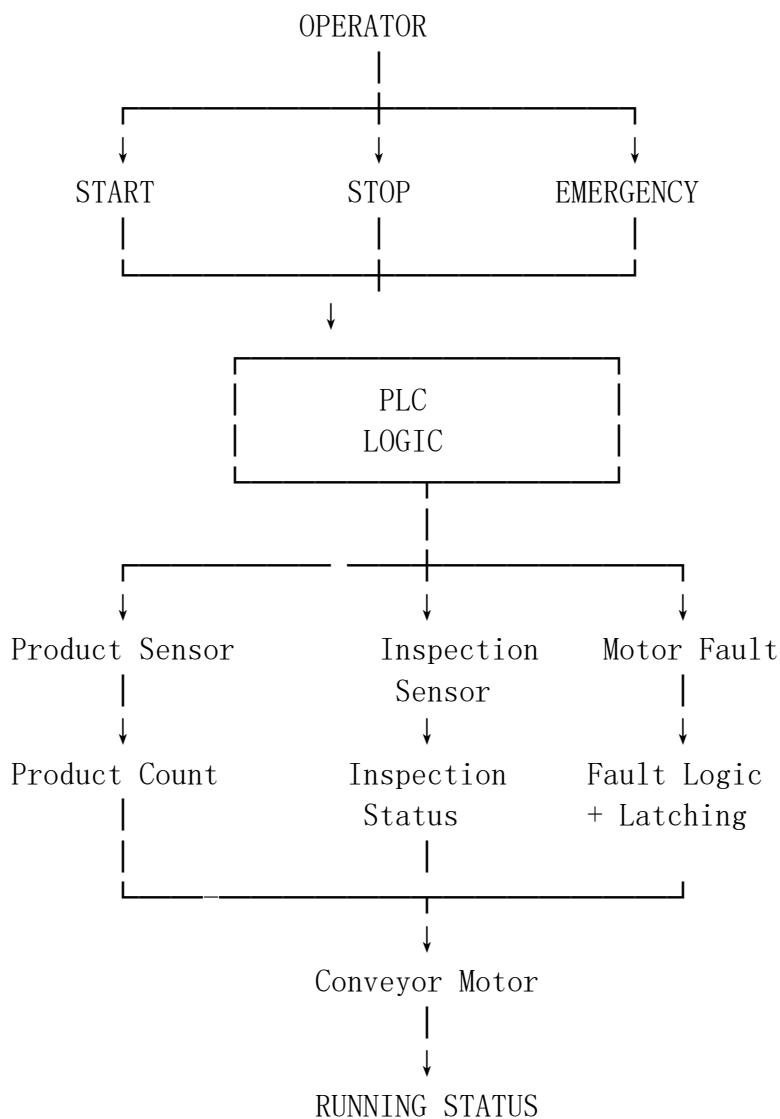
Tools: OpenPLC Editor · Ladder Logic · PLC Simulation

Domain: Industrial Automation / Manufacturing Automation

Project summary:

Designed and simulated a PLC-based automated conveyor control system using OpenPLC and Ladder Diagram programming. The system implements conveyor start/stop control, emergency-stop interlocking, product detection, production counting, inspection-position detection, production-limit control, motor-fault monitoring, fault latching, manual fault reset, and conveyor running-status monitoring.

System architecture :



Final I/O and variable table :

Variable	Type	Role	Purpose
START_BUTTON	BOOL	Input	Starts conveyor
STOP_BUTTON	BOOL	Input	Stops conveyor
EMERGENCY_STOP	BOOL	Safety Input	Simulated emergency-stop interlock
PRODUCT_SENSOR	BOOL	Input	Detects incoming product
COUNTER_RESET	BOOL	Input	Resets production counter
INSPECTION_SENSOR	BOOL	Input	Detects product at inspection position
MOTOR_FAULT	BOOL	Fault Input	Simulates motor fault
FAULT_RESET	BOOL	Input	Clears latched motor fault
CONVEYOR_MOTOR	BOOL	Output/Status	Controls conveyor motor
PRODUCT_DETECTED	BOOL	Status	Indicates product detection
PRODUCT_COUNT	INT	Production Data	Stores product count
PRODUCTION_LIMIT_REACHED	BOOL	Status	Indicates production limit reached
PRODUCT_AT_INSPECTION	BOOL	Status	Indicates product at inspection
MOTOR_FAULT_ACTIVE	BOOL	Fault Status	Indicates active motor fault
FAULT_LATCHED	BOOL	Fault Status	Keeps fault active until reset
CONVEYOR_RUNNING	BOOL	Status	Indicates conveyor is running

Production counter limit: 100

Main functionality :

Conveyor control

The conveyor uses a start/stop seal-in circuit.

Once started, the motor remains ON until a stop condition occurs.

Safety interlock

The simulated emergency-stop input interrupts conveyor operation.

Product detection

The product sensor activates PRODUCT_DETECTED.

Production counting

Each product sensor pulse increments the CTU counter.

Production limit

When the configured production limit is reached:

PRODUCTION_LIMIT_REACHED = TRUE

The conveyor is prevented from continuing.

Inspection position

The inspection sensor produces:

PRODUCT_AT_INSPECTION = TRUE

when a product reaches the inspection position.

Motor fault

A simulated motor fault activates:

MOTOR_FAULT_ACTIVE

and stops the conveyor.

Fault latching

The fault is stored in:

FAULT_LATCHED

Clearing the original motor fault **does not automatically restart the conveyor.**

Manual fault reset

The operator must activate:

FAULT_RESET

to clear the latched fault.

Afterward, the operator must deliberately press START to restart the conveyor.

Final testing documentation :

Test	Expected Result	Result
Start conveyor	Motor turns ON	✓ PASS
Stop conveyor	Motor turns OFF	✓ PASS
Emergency stop	Conveyor operation interrupted	✓ PASS
Product detection	Product status becomes TRUE	✓ PASS
Product counting	Count increments	✓ PASS
Counter reset	Count returns to 0	✓ PASS
Production limit	Limit status activates	✓ PASS
Production-limit interlock	Conveyor prevented from running	✓ PASS
Inspection detection	Inspection status activates	✓ PASS
Motor fault	Motor stops and fault activates	✓ PASS
Fault latching	Fault remains latched	✓ PASS
Clear motor fault	Fault remains latched	✓ PASS
Fault reset	Latch clears	✓ PASS
Deliberate restart	START required to restart	✓ PASS
Running status	Status follows motor state	✓ PASS

Technical description for your portfolio :

PLC-Based Automated Conveyor Control System

Designed and simulated an automated conveyor control system using OpenPLC and Ladder Diagram programming. The system models a basic manufacturing conveyor with operator controls, product detection, production counting, inspection-position detection, production-limit control, motor-fault monitoring, and conveyor status monitoring.

The control logic includes start/stop seal-in functionality, simulated emergency-stop interlocking, automatic production-limit shutdown, and a latched motor-fault mechanism requiring manual fault reset before deliberate conveyor restart.

The project was developed as a PLC simulation to understand fundamental industrial automation concepts including ladder logic, digital inputs and outputs, timers/counters, interlocks, fault handling, and manufacturing-process control.

Screenshots:

SmartFactory_Conveyor

Functions

Function Blocks

Programs

main

Data Types

Global Variables

Resource

Device

Configuration

Servers

Remote Devices

Library

Additional Function Blocks

Standard Function Blocks

Standard Functions

OSCAT Basic

PLCopen SoftMotion (Light)

Manage libraries...

No file selected

main

SmartFactory_ConveyorProgrammain

Description :

Class Filter : All

#

Name

Class

Flags

Type

Location

Initial Value

Documentation

Debug

0

PRODUCT_COUNT

Local

INT

0

1

START_BUTTON

Local

BOOL

TRUE

PRODUCT COUNTING

PRODUCT_SE

INSOR

CTU0

CU

Q

COUNTER_RESET

100

PV

PRODUCTIO

N LIMIT REA

PRODUCT_CO

UNIT

Conveyor_Running_Status

CONVEYOR

MOTOR

CONVEYOR

RUNNING

Console

Debugger

Variables

main.PRODUCT_COUNTINT 0

main.START_BUTTONBOOL TRUE

main.STOP_BUTTONBOOL FALSE

main.CONVEYOR_MOTORBOOL TRUE

main.EMERGENCY_STOPBOOL FALSE

main.PRODUCT_SENSORBOOL FALSE

main.PRODUCT_DETECTEDBOOL FALSE

main.CTUDCTU

main.COUNTER_RESETSHOW HIDE

Range10 seconds

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Class Filter : All

#

Name

Class

Flags

Type

Location

Initial Value

Documentation

Debug

0

PRODUCT_COUNT

Local

INT

0

1

START_BUTTON

Local

BOOL

TRUE

Inspection_Position_Detection

INSPECTION

SENSOR

PRODUCT AT

_INSPECTION

Motor_Fault_Detection

MOTOR_FAULT

UT

MOTOR_FAULT

UT_ACTIVE

Console

Debugger

Variables

main.PRODUCT_COUNTINT 0

main.START_BUTTONBOOL TRUE

main.STOP_BUTTONBOOL FALSE

main.CONVEYOR_MOTORBOOL TRUE

main.EMERGENCY_STOPBOOL FALSE

main.PRODUCT_SENSORBOOL FALSE

main.PRODUCT_DETECTEDBOOL FALSE

main.CTUDCTU

main.COUNTER_RESETSHOW HIDE

Range10 seconds

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Manage libraries...

No file selected

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SmartFactory_Conveyor

Program

main

Description :

Class Filter : All

#	Name	Class	Flags	Type	Location	Initial Value	Documentation	Debug
0	PRODUCT_COUNT	Local		INT		0		
1	START_BUTTON	Local		BOOL		TRUE		

Conveyor_Start_Stop

Product_Detection

Console

Debugger

Variables

main.PRODUCT_COUNT

main.START_BUTTON

main.STOP_BUTTON

main.CONVEYOR_MOTOR

main.EMERGENCY_STOP

main.PRODUCT_SENSOR

main.PRODUCT_DETECTED

main.CTUD

main.PRINTED_BYTE

INT 0

BOOL TRUE

BOOL FALSE

BOOL TRUE

BOOL FALSE

BOOL FALSE

BOOL FALSE

CTU 1

BYTE FALSE

Range 10 seconds

SmartFactory_Conveyor

SmartFactory_Conveyor

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SmartFactory_Conveyor

Program

main

Description :

Class Filter : All

#	Name	Class	Flags	Type	Location	Initial Value	Documentation	Debug
0	PRODUCT_COUNT	Local		INT		0		
1	START_BUTTON	Local		BOOL		TRUE		

Fault_Latch

Fault_Reset

Console

Debugger

Variables

main.PRODUCT_COUNT

main.START_BUTTON

main.STOP_BUTTON

main.CONVEYOR_MOTOR

main.EMERGENCY_STOP

main.PRODUCT_SENSOR

main.PRODUCT_DETECTED

main.CTUD

main.PRINTED_BYTE

INT 0

BOOL TRUE

BOOL FALSE

BOOL TRUE

BOOL FALSE

BOOL FALSE

BOOL FALSE

CTU 1

BYTE FALSE

Range 10 seconds

